

Model Mis-specification Monte Carlo Simulation

Jianing Li

2026-01-07

Project Overview

This document presents a Monte Carlo simulation study examining the finite-sample bias, variance, and MSE of OLS estimators under model misspecification. The simulation focuses on omitted-variable bias under alternative correlation structures and sample sizes. All simulations are implemented in R and are fully reproducible.

Note: This simulation was originally developed as part of a graduate econometrics course and has been lightly reorganized for clarity.

Model mis-specification Monte Carlo simulations

True Model: $y_t = \alpha + \beta x_{1t} + \gamma x_{2t} + u_t$

Under-specified Model: $y_t = \alpha + \beta x_{1t} + e_t$ for $t = 1, 2, \dots, 621$.

(1) Generate the data under the true model

```
library(MASS) # For multivariate models
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
library(reshape2)
```

```
## Warning: package 'reshape2' was built under R version 4.5.2
```

```
# Set seed so Professor Xu can reproduce the same thing :D
set.seed(622)
```

```
n = 621
mu = matrix(c(0, 1), nrow = 2, byrow = TRUE)

sigma1 = 1
sigma2 = 4
alpha = 0.6
beta = 0.2
```

```

gamma = 0.1
rho = 0.126

Sigma = matrix(c(sigma1^2, rho*sigma1*sigma2, rho*sigma1*sigma2, sigma2^2), nrow = 2, byrow = TRUE)

u = rnorm(n, 0, 1)

X = mvrnorm(n, mu = c(0,1), Sigma = Sigma)
x1 = X[, 1]
x2 = X[, 2]

y = alpha + beta*x1 + gamma*x2 + u

```

(2) Run regressions

1. True model

```

true_model = lm(y ~ x1 + x2)
true_model_results = coef(true_model)
true_model_results

```

```

## (Intercept)          x1          x2
##  0.62875826  0.22484298  0.07734491

```

2. Under-specified model

```

under_model = lm(y ~ x1)
under_model_results = coef(under_model)
under_model_results

```

```

## (Intercept)          x1
##  0.6935326  0.2614767

```

(3) Repeat 621 times and Calculate the mean, bias, variance, minimum, maximum and mean square error

```

stats_calculation = function(est, value){
  bias = colMeans(est) - value
  variance = apply(est, 2, var)
  mse = bias ^ 2 + variance

results = data.frame(
  Parameter = colnames(est),
  True_Value = value,
  Mean = colMeans(est),
  Bias = bias,
  Variance = variance,
  MSE = mse,

```

```

    Min = apply(est, 2, min),
    Max = apply(est, 2, max)
)

return(results)
}

simulation = function(rho_value, n_obs, n_sims) {
  n = n_obs
  mu = matrix(c(0, 1), nrow = 2, byrow = TRUE)
  sigma1 = 1
  sigma2 = 4
  alpha = 0.6
  beta = 0.2
  gamma = 0.1

  Sigma = matrix(c(sigma1^2, rho*sigma1*sigma2,
                    rho_value*sigma1*sigma2, sigma2^2), nrow = 2,
                  byrow = TRUE)

  true_model_results = matrix(NA, nrow = n_sims, ncol = 3)
  colnames(true_model_results) <- c("alpha_hat", "beta_hat", "gamma_hat")

  under_model_results = matrix(NA, nrow = n_sims, ncol = 2)
  colnames(under_model_results) = c("alpha_hat", "beta_hat")

  for (i in 1:n_sims) {
    set.seed(i)
    u = rnorm(n, 0, 1)

    X = mvnrm(n, mu = c(0,1), Sigma = Sigma)
    x1 = X[, 1]
    x2 = X[, 2]
    y = alpha + beta*x1 + gamma*x2 + u

    true_model = lm(y ~ x1 + x2)
    true_model_results[i, ] = coef(true_model)

    under_model = lm(y ~ x1)
    under_model_results[i, ] = coef(under_model)
  }

  true_summary = stats_calculation(true_model_results,
                                   c(alpha, beta,
                                     gamma))

  under_summary =
    stats_calculation(under_model_results,
                      c(alpha, beta))

  return(list(
    true_model = true_model_results,
    under_model = under_model_results,

```

```

    true_summary = true_summary,
    under_summary = under_summary,
    rho = rho_value
  ))
}

create_summary_table = function(results, model_name) {
  true_df = results$true_summary
  under_df = results$under_summary

  combined <- rbind(
    cbind(Model = "True", true_df),
    cbind(Model = "Under-specified",
          rbind(under_df,
                data.frame(Parameter = "gamma_hat", True_Value = 0.1,
                          Mean = NA, Bias = NA, Variance = NA, MSE = NA,
                          Min = NA, Max = NA)))
  )

  formatted = combined[, c("Model", "Parameter", "True_Value", "Mean", "Bias", "Variance", "MSE")]
  colnames(formatted) = c("Model", "Parameter", "True Value", "Mean", "Bias", "Variance", "MSE")

  formatted[, 3:7] = round(formatted[, 3:7], 4)

  cat(paste("\n", model_name, "\n"))
  cat("=====\n")
  print(formatted, row.names = FALSE)
  cat("\n")

  return(formatted)
}

results_rho1 = simulation(rho_value = 0.126, n_obs = 621, n_sims = 621)

table_rho1 = create_summary_table(results_rho1, "SUMMARY TABLE: rho = 0.126")

```

```

##
## SUMMARY TABLE: rho = 0.126
## =====
##           Model Parameter True Value   Mean   Bias Variance   MSE
##           True alpha_hat    0.6 0.6031  0.0031  0.0019 0.0019
##           True  beta_hat    0.2 0.1980 -0.0020  0.0016 0.0016
##           True gamma_hat    0.1 0.0999 -0.0001  0.0001 0.0001
## Under-specified alpha_hat    0.6 0.7028  0.1028  0.0020 0.0126
## Under-specified  beta_hat    0.2 0.2501  0.0501  0.0019 0.0044
## Under-specified gamma_hat    0.1    NA    NA    NA    NA

```

```
table_rho1
```

```

##           Model Parameter True Value   Mean   Bias Variance   MSE
## alpha_hat    True alpha_hat    0.6 0.6031  0.0031  0.0019 0.0019
## beta_hat     True  beta_hat    0.2 0.1980 -0.0020  0.0016 0.0016
## gamma_hat    True gamma_hat    0.1 0.0999 -0.0001  0.0001 0.0001

```

```
## alpha_hat1 Under-specified alpha_hat      0.6 0.7028  0.1028   0.0020 0.0126
## beta_hat1  Under-specified beta_hat        0.2 0.2501  0.0501   0.0019 0.0044
## 1          Under-specified gamma_hat       0.1    NA      NA      NA      NA
```

```
create_comparison_plots = function(results, rho_value) {
  true_beta = data.frame(Value = results$true_model[, "beta_hat"], Model = "True Model")
  under_beta = data.frame(Value = results$under_model[, "beta_hat"], Model = "Under-specified")
  beta_data = rbind(true_beta, under_beta)

  true_alpha = data.frame(Value = results$true_model[, "alpha_hat"], Model = "True Model")
  under_alpha = data.frame(Value = results$under_model[, "alpha_hat"], Model = "Under-specified")
  alpha_data = rbind(true_alpha, under_alpha)

  p1 = ggplot(beta_data, aes(x = Value, fill = Model)) +
    geom_histogram(alpha = 0.7, position = "identity", bins = 30) +
    geom_vline(xintercept = 0.2, linetype = "dashed", color = "black", size = 1) +
    labs(title = paste("Distribution of Beta Estimates (rho =", rho_value, ")"),
         x = "Beta Estimate", y = "Frequency") +
    theme_minimal() +
    scale_fill_manual(values = c("True Model" = "blue", "Under-specified" = "red"))

  p2 = ggplot(alpha_data, aes(x = Value, fill = Model)) +
    geom_histogram(alpha = 0.7, position = "identity", bins = 30) +
    geom_vline(xintercept = 0.6, linetype = "dashed", color = "black", size = 1) +
    labs(title = paste("Distribution of Alpha Estimates (rho =", rho_value, ")"),
         x = "Alpha Estimate", y = "Frequency") +
    theme_minimal() +
    scale_fill_manual(values = c("True Model" = "blue", "Under-specified" = "red"))

  gamma_data = data.frame(Value = results$true_model[, "gamma_hat"])
  p3 <- ggplot(gamma_data, aes(x = Value)) +
    geom_histogram(fill = "green", alpha = 0.7, bins = 30) +
    geom_vline(xintercept = 0.1, linetype = "dashed", color = "black", size = 1) +
    labs(title = paste("Distribution of Gamma Estimates (rho =", rho_value, ")"),
         x = "Gamma Estimate", y = "Frequency") +
    theme_minimal()

  bias_data = data.frame(
    Model = c("True", "Under-specified"),
    Beta_Bias = c(results$true_summary$Bias[2], results$under_summary$Bias[2]),
    Alpha_Bias = c(results$true_summary$Bias[1], results$under_summary$Bias[1])
  )

  bias_melted = melt(bias_data, id.vars = "Model")
  p4 = ggplot(bias_melted, aes(x = Model, y = value, fill = variable)) +
    geom_bar(stat = "identity", position = "dodge") +
    geom_hline(yintercept = 0, linetype = "dashed", color = "black") +
    labs(title = paste("Bias Comparison (rho =", rho_value, ")"),
         y = "Bias", fill = "Parameter") +
    theme_minimal() +
    scale_fill_manual(values = c("Beta_Bias" = "darkblue", "Alpha_Bias" = "darkred"))

  print(p1)
  print(p2)
```

```

print(p3)
print(p4)

return(list(p1, p2, p3, p4))
}

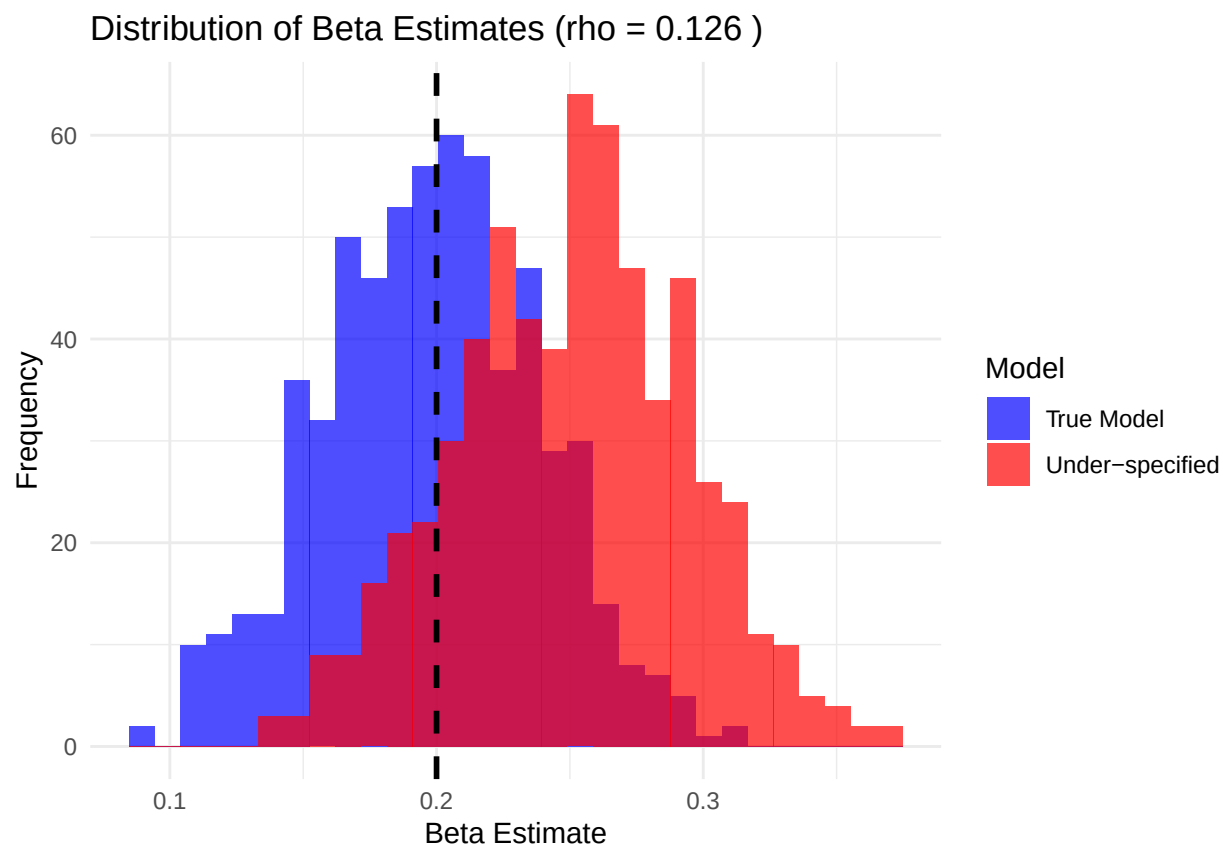
plots_rho1 = create_comparison_plots(results_rho1, 0.126)

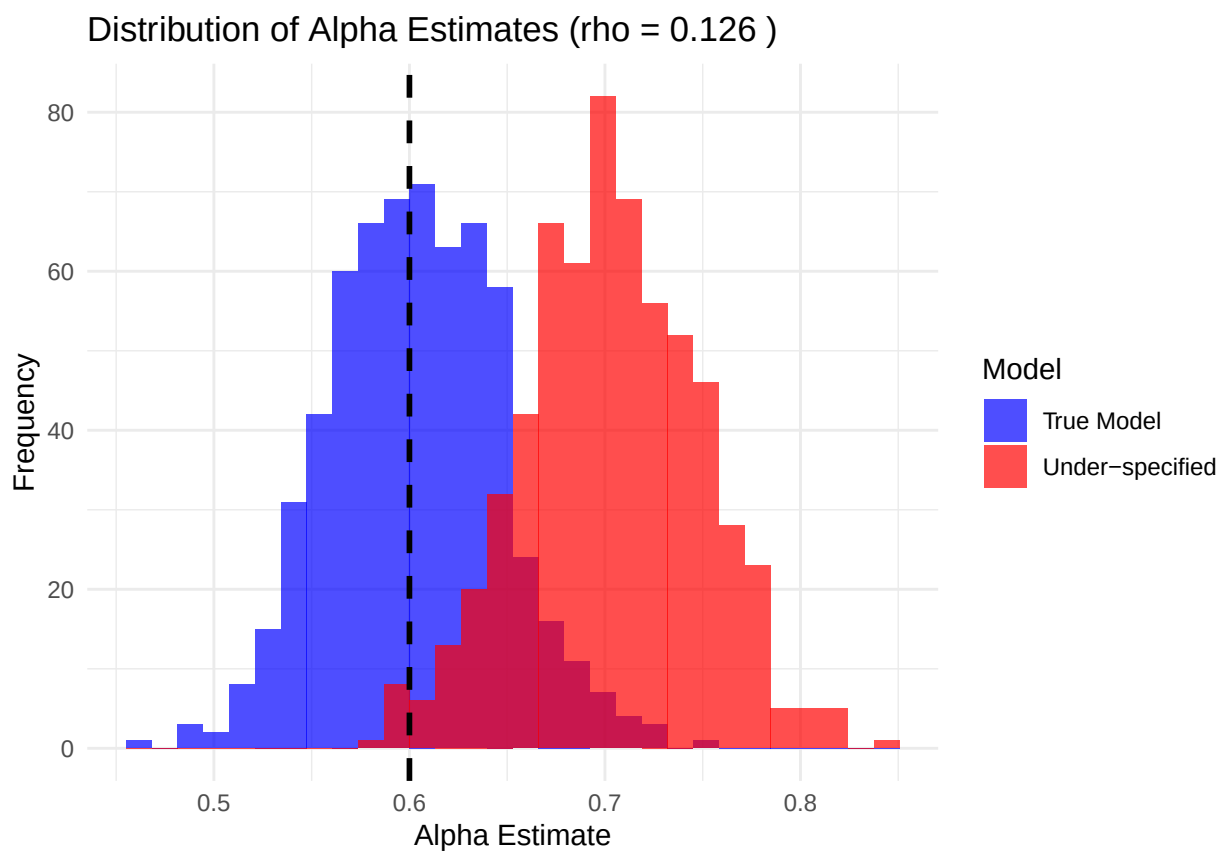
```

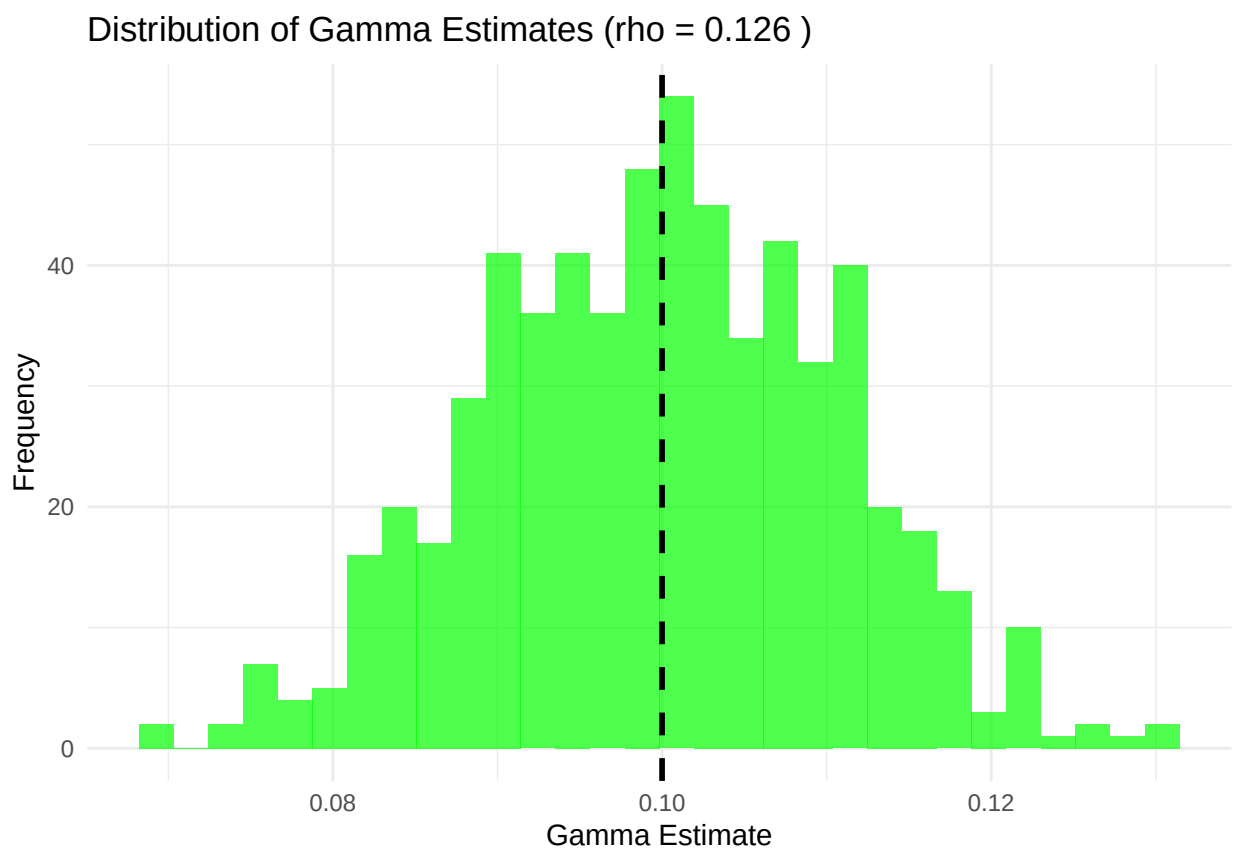
```

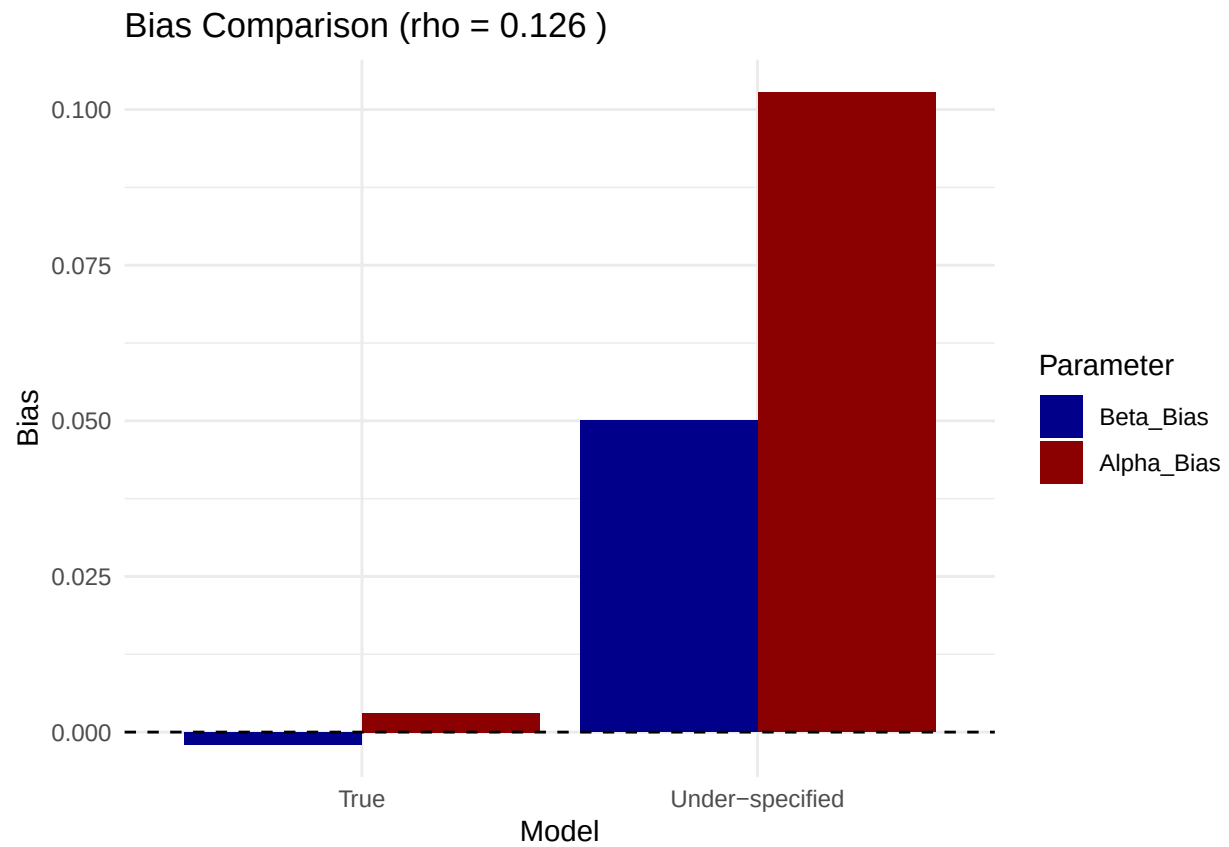
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

```



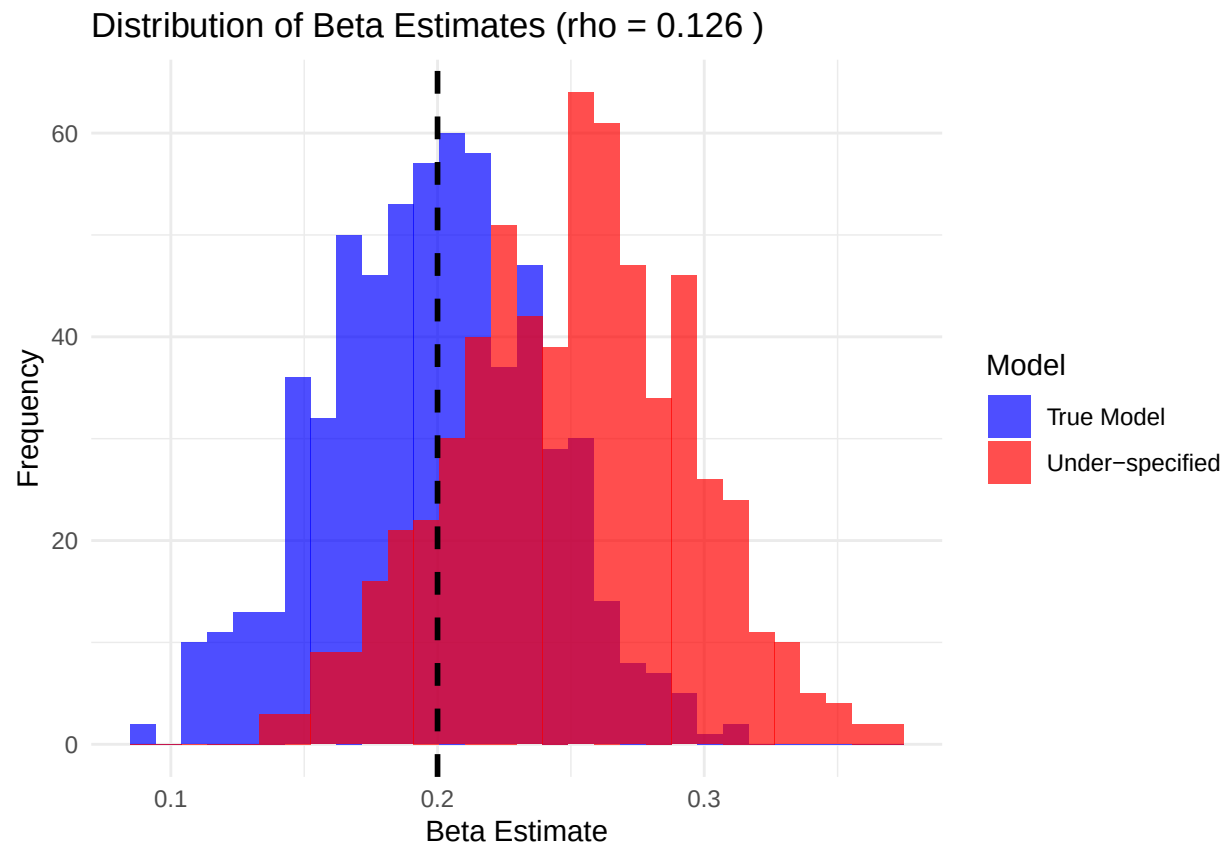




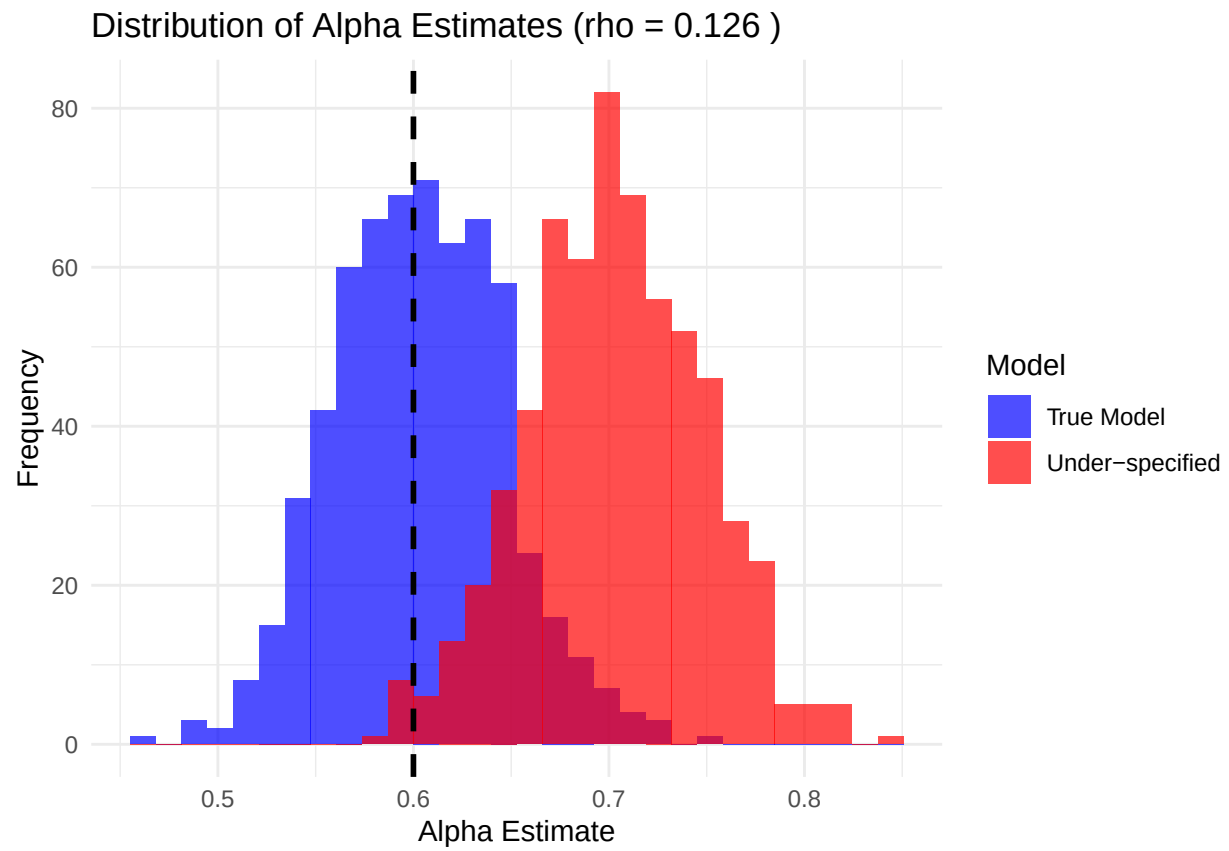


```
plots_rho1
```

```
## [[1]]
```

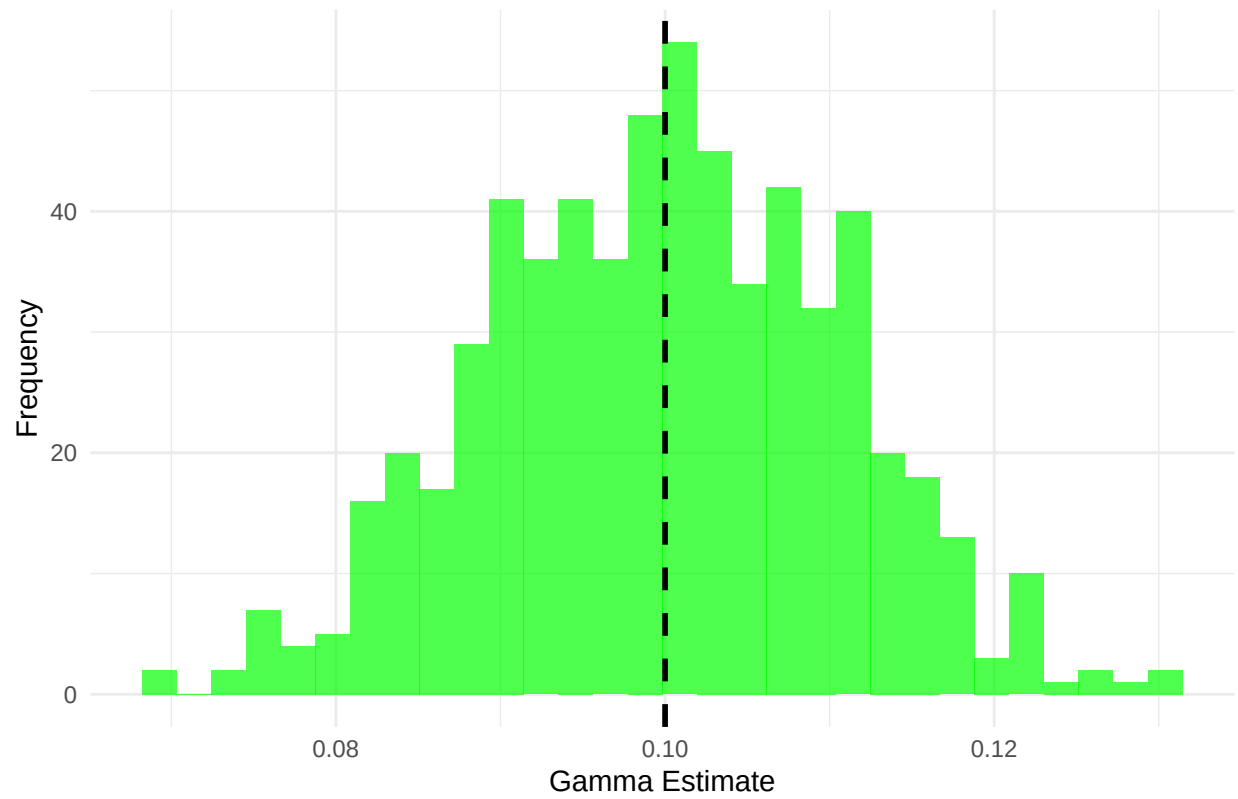


```
##  
## [[2]]
```

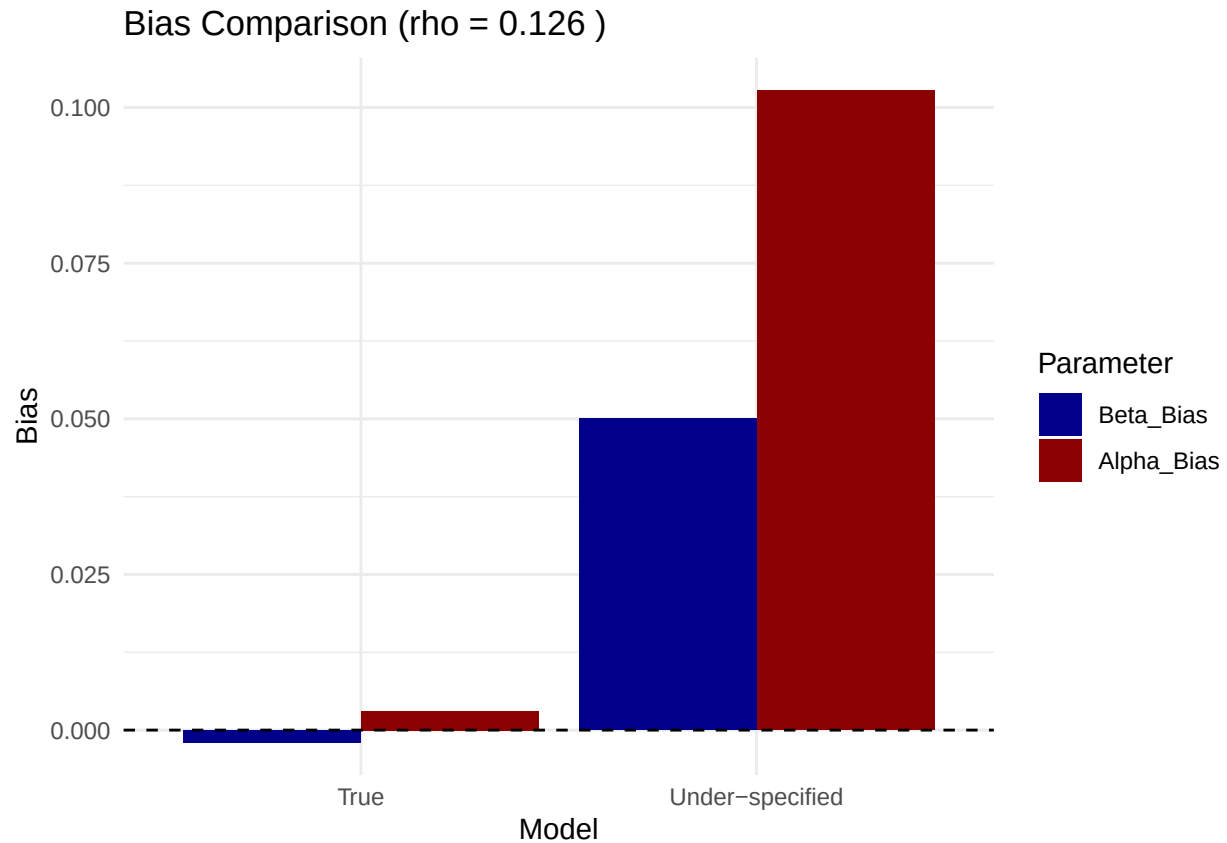


```
##  
## [[3]]
```

Distribution of Gamma Estimates (rho = 0.126)



```
##  
## [[4]]
```

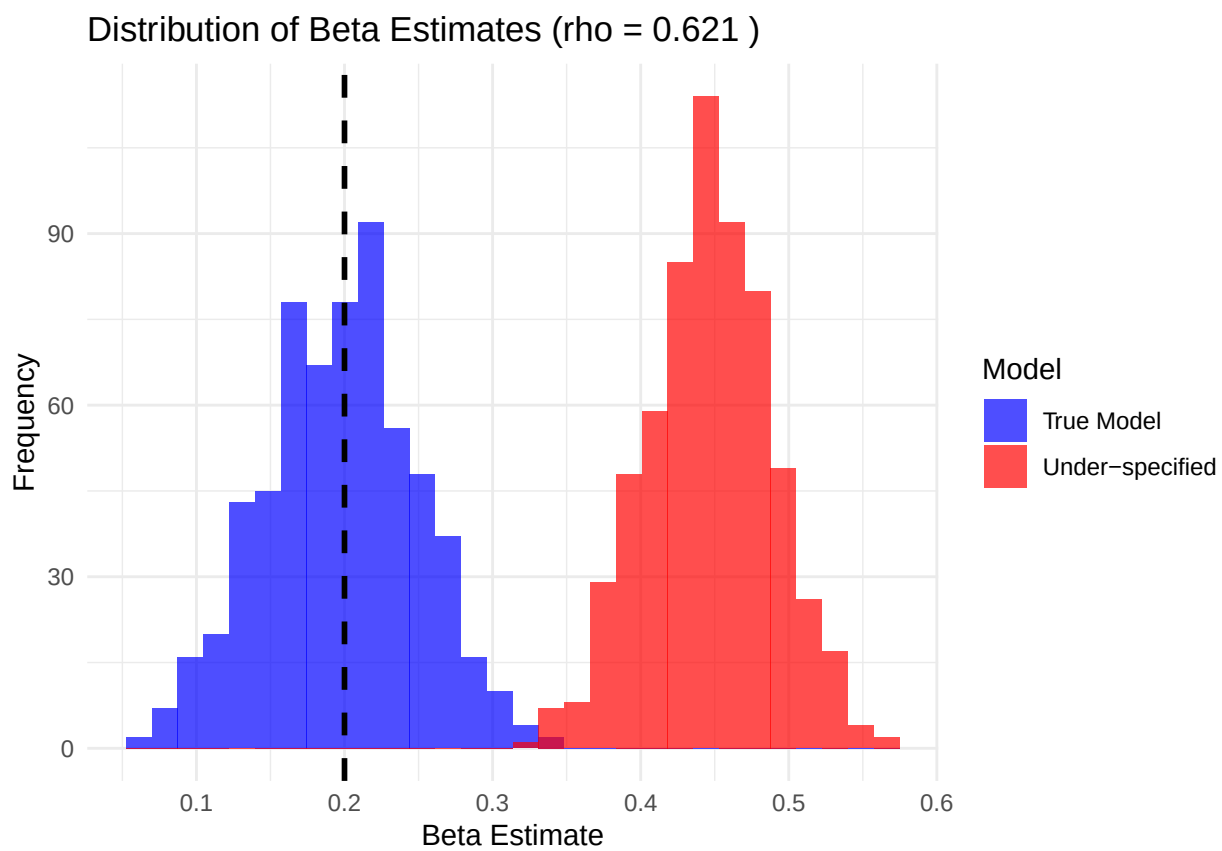


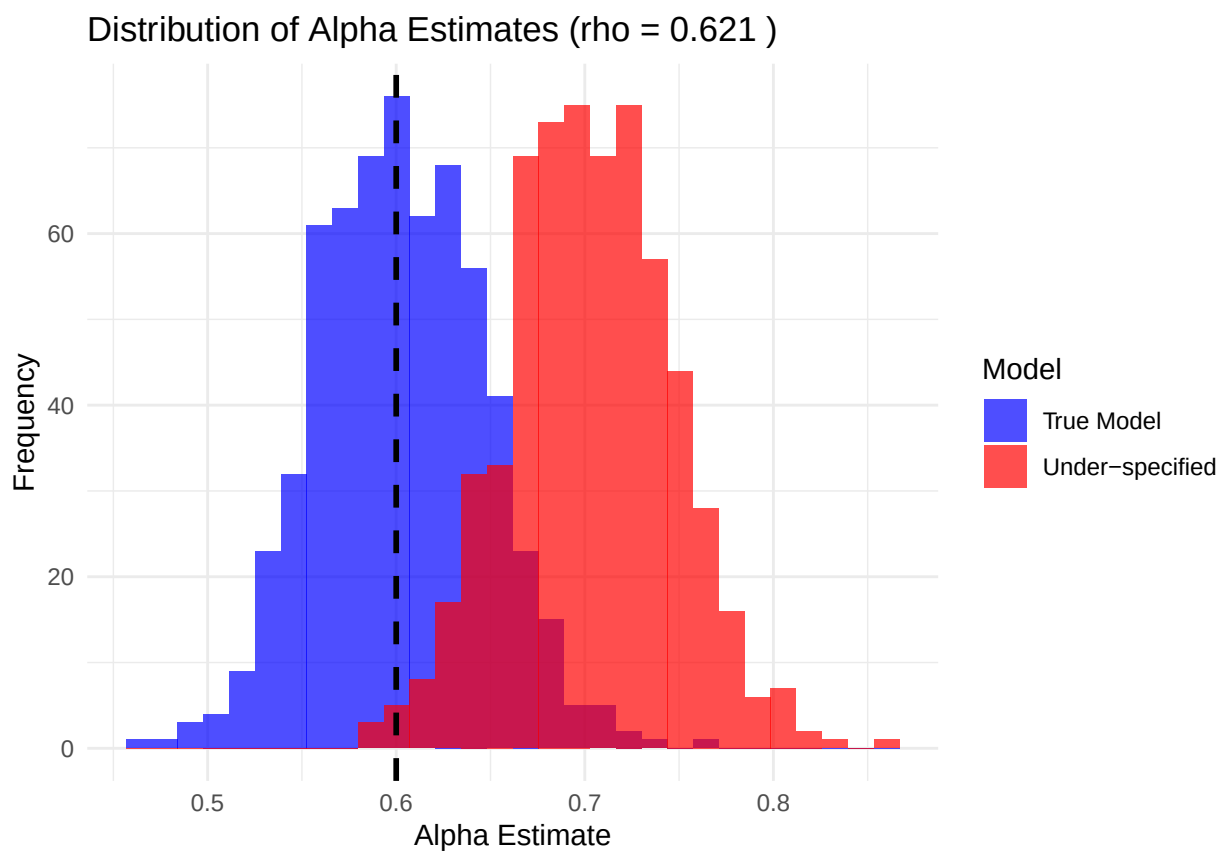
(4) Repeat 621 times with $\rho = 0.621$

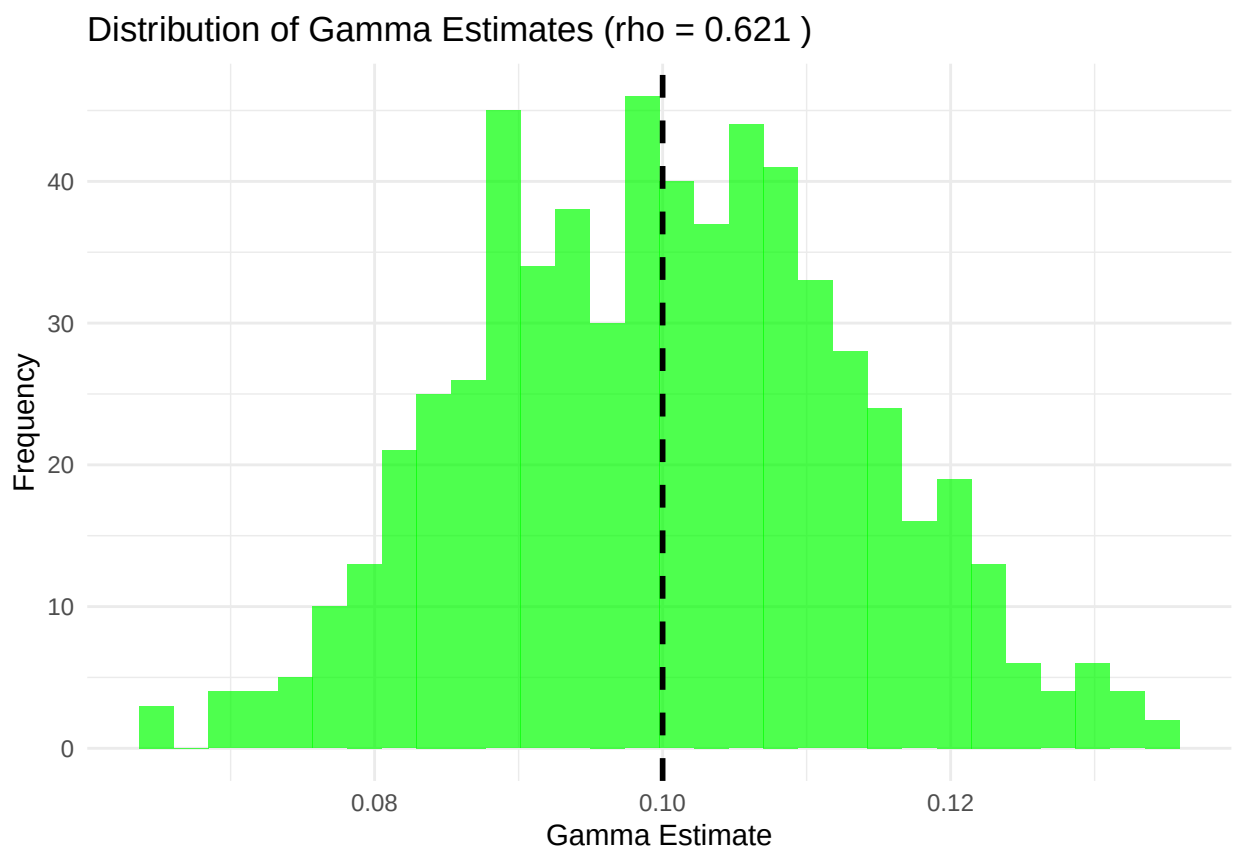
```
results_rho2 = simulation(rho_value = 0.621, n_obs = 621, n_sims = 621)
table_rho2 = create_summary_table(results_rho2, "SUMMARY TABLE: rho = 0.621")
```

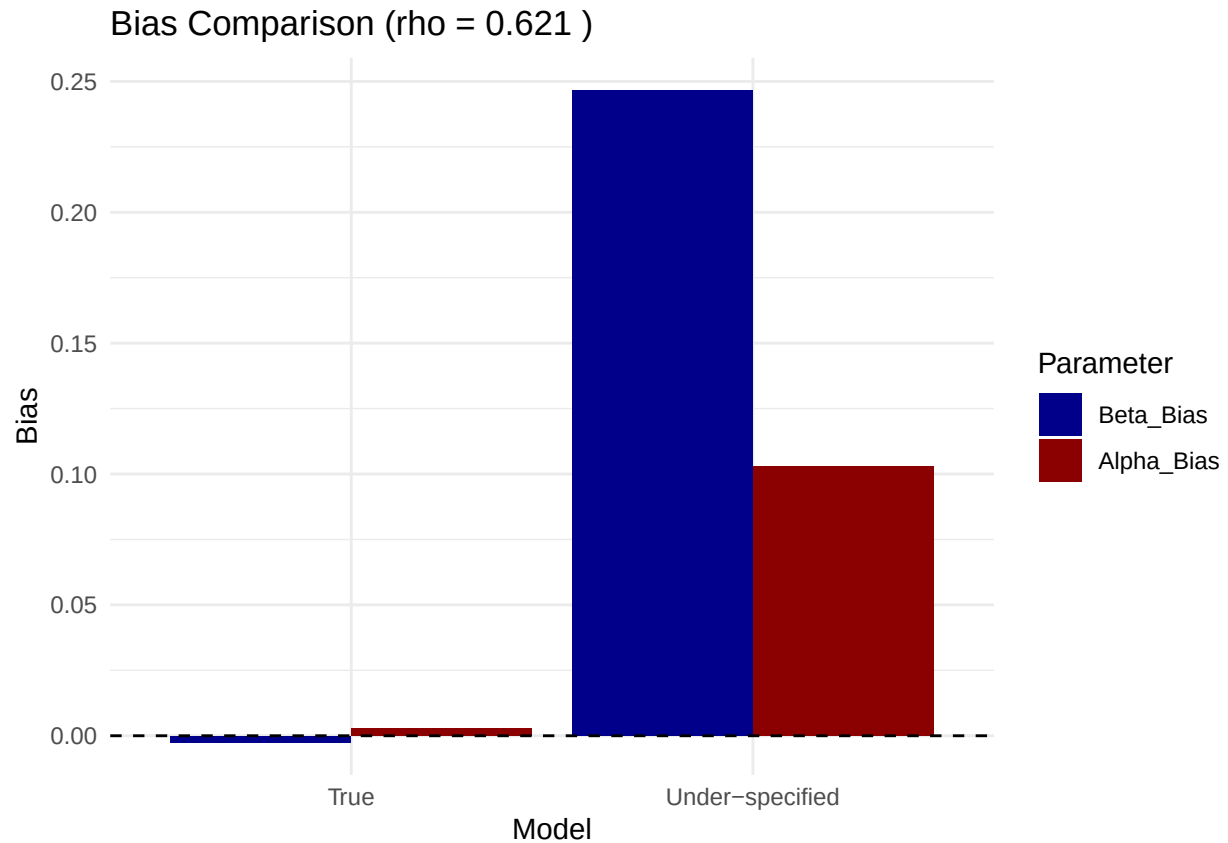
```
##
## SUMMARY TABLE: rho = 0.621
## =====
##           Model Parameter True Value   Mean   Bias Variance   MSE
##           True  alpha_hat    0.6 0.6027  0.0027  0.0019 0.0019
##           True  beta_hat     0.2 0.1975 -0.0025  0.0026 0.0026
##           True  gamma_hat    0.1 0.1002  0.0002  0.0002 0.0002
## Under-specified alpha_hat    0.6 0.7028  0.1028  0.0019 0.0124
## Under-specified beta_hat     0.2 0.4466  0.2466  0.0018 0.0626
## Under-specified gamma_hat    0.1    NA     NA     NA     NA
```

```
plots_rho2 = create_comparison_plots(results_rho2, 0.621)
```









(5) Comments

Comparing to the true model, under-specified model shows a significant bias in β estimates. When $\rho = 0.126$ is lower, it produces small bias, when $\rho = 0.621$ is higher, it gives larger bias, which results in a poor performance.

From above, we assume that the cost of omitting variables increases with their correlation to included variables. High correlation makes misspecification worsen in bias and performance.